

Assignment, Basic Algorithms and C

Statements with solutions (algorithm + C)

Exercise 1 — Simple display

Task. Write a program that prints Hello!.

Exercise 2 — Assignment and sum

Task. Read two integers a and b , compute $s = a + b$, display s .

Exercise 3 — Swapping variables

Task. Swap the values of two integers x and y .

Exercise 4 — Average of three numbers

Task. Read real numbers a, b, c , display their average.

Exercise 5 — Celsius→Fahrenheit conversion

Task. Read C in °C, display $F = 1.8C + 32$.

Exercise 6 — Maximum of two integers

Task. Read a, b , display the larger of the two.

Exercise 7 — Maximum of three numbers

Task. Read x, y, z and display the maximum.

Exercise 8 — Even or odd

Task. Read an integer n , display `even` or `odd`.

Exercise 9 — Absolute value

Task. Read a real number x , display $|x|$.

Exercise 10 — Roots of a quadratic

Task. Read a, b, c , compute the discriminant $\Delta = b^2 - 4ac$ and display the roots according to the sign of Δ .

Exercise 11 — Multiplication table

Task. Read n and display its multiplication table from 1 to 10.

Exercise 12 — Factorial (iterative)

Task. Read $n \geq 0$, compute $n!$.

Exercise 13 — Fibonacci sequence

Task. Read n , display the first n Fibonacci terms (starting with 0, 1).

Exercise 14 — Sum from 1 to n

Task. Read n , compute $S = \sum_{k=1}^n k$.

Exercise 15 — Count digits of an integer

Task. Read an integer (absolute value), display its number of digits.

Exercise 16 — Reverse digits

Task. Read a positive integer, display the integer obtained by reversing the order of its digits.

Exercise 17 — Primality test (naive)

Task. Read $n \geq 2$, state whether it is prime.

Exercise 18 — GCD (Euclid's algorithm)

Task. Read $a, b > 0$, display $\text{GCD}(a, b)$.

Exercise 19 — Min, max and average of an array

Task. Read n ($1 \leq n \leq 100$) real values into an array, display min, max and average.

Exercise 20 — Counting vowels (string)

Task. Read a line of text, count the number of vowels (a, e, i, o, u, y), ignoring case.